

Valuing American Options

Using Least-Squares Policy Iteration

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Abstract

This project investigates the valuation of one-dimensional American call and put options using Least-Squares Policy Iteration (LSPI). The valuation problem is formulated as an optimal stopping problem and represented as a Markov decision process, where the holder chooses between exercising the option and continuing to hold it. LSPI is used to approximate the optimal exercise policy and corresponding option value. The performance of the method is evaluated numerically by comparing its option values and exercise strategies with those obtained using an implicit finite-difference method.

1 Introduction

Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined strike price [2]. European options may only be exercised at maturity, whereas American options may be exercised at any time before or at maturity. This additional exercise flexibility makes the valuation of American options an optimal stopping problem, since the holder must determine whether to exercise immediately or continue holding the option.

This project investigates the valuation of one-dimensional American call and put options using LSPI, which was first devised in [3]. The option valuation problem is formulated as a Markov decision process, where the underlying asset price represents the state and the decision to exercise or continue represents the action. LSPI is then used to approximate the optimal exercise policy and the corresponding option value.

The performance of LSPI is evaluated numerically by comparing the resulting option values and exercise strategies with those obtained using an implicit finite-difference method. The aim is to investigate the effectiveness of LSPI in approximating the optimal exercise policy and valuing American options.

2 Mathematical Background

Let S_t denote the price of the underlying asset at time t , where $t \in [0, T]$ and T is the maturity time. Let X denote the strike price, r the continuously compounded risk-free interest rate, q the continuous dividend yield, and σ the volatility of the underlying asset. We denote by $W_t^{\mathbb{Q}}$ a standard Brownian motion under the risk-neutral measure $\mathbb{Q}$.

2.1 Optimal Stopping Problem

For a European option, the exercise time is fixed at maturity. In contrast, an American option may be exercised at any stopping time $\tau \in [0, T]$. This flexibility makes the valuation of an American option an optimal stopping problem.

For an American option with payoff function g , its value u at time t when the underlying asset price is S_t is

$$u(t, S_t) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau-t)} g(S_\tau) \mid S_t \right], \quad (2.1)$$

where $\mathcal{T}_{t,T}$ denotes the set of stopping times taking values between t and T .

For an American put and call, the payoff functions are respectively

$$g_{\text{put}}(S) = \max(X - S, 0), \quad g_{\text{call}}(S) = \max(S - X, 0).$$

The optimal stopping problem can be expressed through the dynamic programming principle [1, 4, 5]. For a sufficiently small time step Δt ,

$$u(t, S) = \max \left\{ g(S), e^{-r\Delta t} \mathbb{E}^{\mathbb{Q}} [u(t + \Delta t, S_{t+\Delta t}) \mid S_t = S] \right\}.$$

The first term represents the value of exercising immediately, while the second represents the value of continuing to hold the option. Thus, the optimal policy is determined by comparing the immediate exercise value with the continuation value.

2.2 Variational Inequality

Under the risk-neutral measure $\mathbb{Q}$, the underlying asset price satisfies

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (2.2)$$

Let $u(t, S)$ denote the value of the option. In the continuation region, where it is optimal not to exercise, the discounted option value must be a martingale under the risk-neutral measure. Therefore,

$$e^{-rt} u(t, S_t)$$

is a martingale.

Applying Itô's lemma to $u(t, S_t)$ gives

$$du = \left(\frac{\partial u}{\partial t} + (r - q)S_t \frac{\partial u}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 u}{\partial S^2} \right) dt + \sigma S_t \frac{\partial u}{\partial S} dW_t^{\mathbb{Q}}.$$

Using the product rule,

$$d \left(e^{-rt} u(t, S_t) \right) = e^{-rt} du - r e^{-rt} u(t, S_t) dt.$$

Substituting the expression for du gives

$$\begin{aligned} d \left(e^{-rt} u(t, S_t) \right) &= e^{-rt} \left[\frac{\partial u}{\partial t} + (r - q)S_t \frac{\partial u}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 u}{\partial S^2} - ru \right] dt \\ &\quad + e^{-rt} \sigma S_t \frac{\partial u}{\partial S} dW_t^{\mathbb{Q}}. \end{aligned}$$

Since $e^{-rt} u(t, S_t)$ is a martingale, its drift must vanish. Consequently,

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 u}{\partial S^2} + (r - q)S \frac{\partial u}{\partial S} - ru = 0. \quad (2.3)$$

This is the Black-Scholes partial differential equation in the continuation region.

For an American option, the holder may exercise at any time before or at maturity. Therefore, the option value must satisfy the early-exercise constraint

$$u(t, S) \geq g(S).$$

In the continuation region,

$$u(t, S) > g(S),$$

and the Black-Scholes PDE holds. In the exercise region,

$$u(t, S) = g(S).$$

The two conditions can be combined into the variational inequality

$$\max \left\{ g(S) - u(t, S), \frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 u}{\partial S^2} + (r - q)S \frac{\partial u}{\partial S} - ru \right\} = 0. \quad (2.4)$$

The terminal condition is

$$u(t, S) = g(S).$$

Thus, the valuation of an American option is characterized by a free-boundary problem. The boundary separating the exercise and continuation regions is the optimal exercise boundary.

2.3 Implicit Finite-Difference Scheme

Let

$$S_i = i\Delta S, \quad \tau_n = n\Delta\tau,$$

and denote

$$u_i^n \approx u(\tau_n, S_i).$$

Using backward Euler in τ and central finite differences in S gives

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta\tau} = A\mathbf{u}^{n+1},$$

where A is the tridiagonal finite-difference matrix associated with the operator $\mathcal{L}$. Therefore,

$$(I - \Delta\tau A) \mathbf{u}^{n+1} = \mathbf{u}^n.$$

At each time step, the American early-exercise constraint is imposed by

$$\mathbf{u}^{n+1} \leftarrow \max(\mathbf{u}^{n+1}, \mathbf{g}),$$

where the maximum is taken componentwise and

$$\mathbf{g} = \begin{bmatrix} g(S_1) \\ \vdots \\ g(S_{M-1}) \end{bmatrix}.$$

The initial condition is

$$\mathbf{u}^0 = \mathbf{g}.$$

Thus, the implicit finite-difference method advances the solution forward in the time-to-maturity variable τ , solving a tridiagonal linear system at each time step and enforcing the early-exercise constraint.

3 Least-Squares Policy Iteration

We formulate the American option valuation problem as a finite-horizon Markov decision process. Let

$$t_n = n\Delta t, \quad n = 0, 1, \dots, N,$$

denote the discretised time points, where Δt is the time step and $T = N\Delta t$ is the maturity time. Let $S_n = S_{t_n}$ denote the underlying asset price at time t_n .

The action space is defined by

$$\mathcal{A} = \{\text{exercise, continue}\}.$$

The immediate payoff associated with exercise is denoted by $g(S_n)$, where g is the payoff function defined in Section 2. Let R_n denote the reward obtained at time t_n , and let γ denote the discount factor. A policy is denoted by

$$\pi : \mathcal{S} \rightarrow \mathcal{A},$$

where $\mathcal{S}$ is the state space of the underlying asset price.

The objective of LSPI is to determine an approximately optimal policy for the resulting Markov decision process. The method combines policy iteration with least-squares function approximation to estimate the action-value function.

3.1 Markov Decision Process Formulation

The state at time t_n is given by

$$S_n = S_{t_n}.$$

The holder chooses an action

$$a_n \in \mathcal{A}.$$

If

$$a_n = \text{exercise},$$

the option terminates and the holder receives the immediate payoff

$$R_n = g(S_n).$$

If

$a_n = \text{continue}$,

the option remains active and the underlying asset evolves according to the risk-neutral dynamics specified in Equation (2.2). The corresponding transition is therefore from S_n to the next state S_{n+1} .

The action-value function associated with a policy π is defined by

$$Q^\pi(S, a) = \mathbb{E}^\mathbb{Q} \left[\sum_{k=n}^N \gamma^{k-n} R_k \mid S_n = S, a_n = a, \pi \right]. \quad (3.1)$$

The optimal action-value function is

$$Q^*(S, a) = \sup_{\pi} Q^\pi(S, a). \quad (3.2)$$

The optimal policy is consequently given by

$$\pi^*(S) = \arg \max_{a \in \mathcal{A}} Q^*(S, a). \quad (3.3)$$

3.2 Action-Value Function Approximation

LSPI approximates the action-value function using a linear combination of basis functions. Let

$$\phi(S, a) = \begin{bmatrix} \phi_1(S, a) \\ \vdots \\ \phi_K(S, a) \end{bmatrix}$$

denote the vector of K basis functions. The action-value function is approximated by

$$\hat{Q}(S, a; \mathbf{w}) = \phi(S, a)^\top \mathbf{w}, \quad (3.4)$$

where

$$\mathbf{w} \in \mathbb{R}^K$$

is the vector of unknown approximation coefficients.

Given a collection of transitions

$$(S_k, a_k, R_k, S'_k), \quad k = 1, \dots, M,$$

the Bellman equation for a fixed policy π is approximated by

$$\hat{Q}(S_k, a_k) \approx R_k + \gamma \hat{Q}(S'_k, \pi(S'_k)).$$

Substituting the linear approximation gives

$$\phi(S_k, a_k)^\top \mathbf{w} \approx R_k + \gamma \phi(S'_k, \pi(S'_k))^\top \mathbf{w}.$$

The least-squares approximation is obtained by solving

$$A\mathbf{w} = \mathbf{b}, \quad (3.5)$$

where

$$A = \sum_{k=1}^M \phi(S_k, a_k) [\phi(S_k, a_k) - \gamma \phi(S'_k, \pi(S'_k))]^\top \quad (3.6)$$

and

$$\mathbf{b} = \sum_{k=1}^M \phi(S_k, a_k) R_k. \quad (3.7)$$

3.3 Policy Improvement

After estimating the coefficient vector $\mathbf{w}$, the policy is improved by selecting the action with the largest estimated action value:

$$\pi_{\text{new}}(S) = \arg \max_{a \in \mathcal{A}} \hat{Q}(S, a; \mathbf{w}).$$

For the American option problem, this gives

$$\pi_{\text{new}}(S) = \begin{cases} \text{exercise,} & \hat{Q}(S, \text{exercise}) \geq \hat{Q}(S, \text{continue}), \\ \text{continue,} & \hat{Q}(S, \text{exercise}) < \hat{Q}(S, \text{continue}). \end{cases}$$

The policy evaluation and policy improvement steps are repeated until the policy converges or a prescribed stopping criterion is satisfied. The resulting policy provides an approximation to the optimal exercise strategy, while the corresponding action-value function provides an approximation to the American option price.

4 Numerical Setup

This section specifies the financial and numerical parameters used in the experiments. The same financial parameters are used for both the LSPI and implicit finite-difference methods to ensure a consistent comparison.

4.1 Financial Parameters

The numerical experiments consider one-dimensional American call and put options under the risk-neutral geometric Brownian motion model introduced in Equation (2.2). The financial parameters used in the experiments are given in Table 1.

Table 1: Option contract parameters

Parameter	Value
Initial asset price S_0	10
Strike price X	10
Maturity T	0.75 years
Risk-free interest rate r	0.05
Dividend yield q	0
Volatility σ	0.20

4.2 Simulation of the Underlying Asset

For the LSPI method, the underlying asset price is simulated using the exact transition distribution of the geometric Brownian motion. For $t_{n+1} = t_n + \Delta t$,

$$S_{n+1} = S_n \exp \left[\left(r - q - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_n \right], \quad Z_n \sim \mathcal{N}(0, 1). \quad (4.1)$$

The simulated asset-price paths provide the transitions required by the LSPI algorithm.

4.3 Numerical Parameters

The numerical parameters used by the LSPI and implicit finite-difference methods are given in Table 2. The parameters are chosen to provide a balance between computational cost and numerical accuracy.

Table 2: Numerical parameters used in the experiments.

Parameter	Value
Number of simulated paths N_{paths}	50 000
Number of LSPI time steps N_t	100
Number of basis functions K	5
Number of finite-difference time steps N	200
Number of asset-price grid points M	400
Maximum asset price S_{max}	40
LSPI convergence tolerance	10^{-6}
Maximum number of policy iterations	100

The same financial parameters are used for both methods. The LSPI method uses simulated asset-price paths generated from Equation (4.1), while the implicit finite-difference method uses the asset-price and time-to-maturity discretisation described in Section 2.3. The resulting numerical estimates are compared in terms of option values and the corresponding optimal exercise strategies.

5 Numerical Results

5.1 Option Value Comparison

5.2 Exercise Policy Comparison

5.3 Numerical Error and Convergence

References

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